Rung 0 — measure the error rate before trying to change it

A one-page research protocol for the clinical team (MVA Hackathon 2026, Track 2). It gives no treatment and changes nothing in the child's care; it needs the clinical team's judgement, ethics approval and the family's consent. Code and assumptions: `scripts/t2/nof1.py`*; results:* `reports/t2/nof1.md`*.*

Why. Every candidate in the Track 2 report needs a readout that moves in weeks, and the usual ones do not: a lymphocyte karyotype reports aneuploidy accumulated over years. The frequency of **micronucleated reticulocytes (%MN-RET)** counts micronuclei — the direct product of chromosome missegregation — in red cells that circulate for 1–2 days, so it reports a *recent* error rate [1, 2]. It has a paediatric reference (upper bound of the 90 % tolerance interval in controls, **0.42 %**) [3], it rose within 3–5 days in 9 of 10 adults starting chemo- or radiotherapy [4], and it was 5.2-fold raised in untreated high-grade glioma [7] — but it has **never been measured in MVA**. Two questions first: is it raised in this child, and how much does it vary between visits (σ)? Every n-of-1 design depends on σ.

Item	Specification	Basis
Samples	6 over 8 weeks (weeks 0, 1, 2, 4, 6, 8) plus 1 spare	about 10 % of paediatric samples failed QC [3]
Material	240 µL of whole blood from a K2-EDTA tube, ideally drawn at a routine blood test (no extra venepuncture)	[3]
Processing	fixed in ultracold methanol per kit instructions, stored at −75 to −85 °C, **all samples analysed in one batch**	[3]; batching removes run-to-run variation
Events	≥ 20,000 $CD71^+$ reticulocytes per sample (minimum 10,000); **≥ 100,000 if the baseline is below 1 %**	[3]; below 1 % counting error starts to cost power
Off treatment	*suggested:* no cytotoxic chemotherapy or radiotherapy in the previous 4 weeks (to agree with the clinical team)	the rise peaks 3–5 days after starting therapy, up to 26-fold [4]; time back to baseline is not established
Record each visit	infection (fever, CRP), nutritional status, folate and B12, reticulocyte count, transfusions	infection and malnutrition raise %MN-RET in children [5]; folate depletion raises micronuclei [6]

Pre-declared analysis. Report the geometric-mean %MN-RET with its 90 % interval and $\hat{\sigma}$, the SD of log %MN-RET between visits. **Plan with the upper 80 % bound of $\hat{\sigma}$, not the point estimate**: with six samples $\hat{\sigma}$ is imprecise. The trial itself re-estimates σ (t posterior), but how many visits it needs depends on the true σ.

Result	Reading	Next step
A · mean ≤ 0.42 %	the endpoint does not see this child's instability in blood	do not use it for an n-of-1; report it as the first MVA measurement — it is also evidence against the disease model the measurement plan rests on
B · raised, $\hat{\sigma}$ ≤ 0.5	an n-of-1 on this endpoint is feasible	size it from the table below, reading the column at or above the upper bound of $\hat{\sigma}$
C · raised, $\hat{\sigma}$ > 0.5	too noisy for a decision within weeks	sample more often, or choose another endpoint

If an mTOR inhibitor is ever started (tier 3), the same assay becomes the safety monitor, because rapamycin was aneugenic in human lymphocytes in one in-vitro study [8]: pre-declare a stop rule, e.g. P(increase ≥ 25 %) ≥ 90 %.

Visits per condition (on/off) for 80 % power, 20,000 reticulocytes per sample; each cell gives the visits for a true 35 % / 50 % reduction, «—» = more than 30.

Responder rule (false-positive rate)	σ = 0.20	σ = 0.35	σ = 0.50
reduction ≥ 25 %, 95 % sure (< 1 %)	28 / 5	— / 11	— / 20
any reduction, 95 % sure (4 %)	4 / 3	9 / 5	18 / 8
any reduction, 90 % sure (10 %)	3 / 2	7 / 4	13 / 6

The rule is a choice for the clinical team and the family, **made before the first sample**: a strict rule needs a large effect or many visits, because its threshold sits close to the effect one hopes for; a loose rule accepts more false positives.

What this is not. Not a validated diagnostic test (a first disease-activity observation); not validated in MVA; it reads missegregation in the erythroid lineage, not in muscle or other tissues; a fall in %MN-RET is a pharmacodynamic signal, not proof of clinical benefit.

References (PMID). [1] Dertinger 2003, 14644356 · [2] Dertinger 2011, 21164195 · [3] Baig 2020, 32761646 (paediatric) · [4] Dertinger 2007, 17059891 · [5] Cervantes-Ríos 2012, 22119781 · [6] Everson 1988, 3367390 · [7] Barrera 2018, 30335243 · [8] Bonatti 1998, 9914383.